cloudspaces

Privacy Aware Data Sharing in the Cloud



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Hot topic of the day!

- *Your mails are being read!*
- *Deutsche Telekom proposes that not a single byte of local data should leave German borders.*
- *If successful, then this could be done in entire Schengen area*

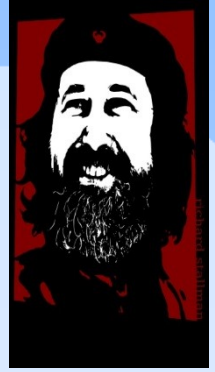
Problem

We would still like to use the Cloud and avail the services provided by cloud providers such as storage, music streaming, online collaboration on documents, etc.

Solution

Become an Information Guerilla?

Legendary Hacker Richard Stallman



- *Only uses search engines on other people's machines*
- *Doesn't buy anything on the Internet*
- *Doesn't use software like Skype*
- *And naturally rejects cloud environments*

What options do PEOPLE have?

- *Can’t be like him*
- *Even if they had the technical expertise and know how*
- *Even then, it’s a lot of effort and time*
- *Can’t realistically expect people to keep everything on their own computer (in this day and age, syncing etc)*
- *Can’t encrypt everything. Then you can lose services. Also many existing platforms don’t provide that*

Some Related Work

- **Online Social Networks:** Pioneering work in quantifying the privacy risk of exposing profile items by Liu and Terzi [1].
- **Ubiquitous Computing:** Pallapa et al. consider each sensing unit and propose ways of making information private based on context [2].
- **File Classification:** In 2011, Microsoft released a report describing its File Classification Infrastructure (FCI), used internally by its IT for obtaining file-level content-based sensitivity
- **Miscellaneous:** *location privacy, db tables etc*

So what is privacy?

- *Its not only content such as pin identifiable numbers, credit card numbers etc.*
- *The “context” in which content is shared is sometimes equally (if not more) important*
- *An example given by Joe Biden, the Vice President of US before he was the VP, about tapping phone calls*
- *Of course now he says the opposite*

Automated Context Aware Crowd Sourced Privacy

Context

- *We propose to take into account the context in which a data item is shared and present people with the risk of sharing*

- *Example of context:*

A powerpoint presentation authored by me in my work computer shared with my colleague

as opposed to the same data item which I share with my friend who is not a colleague

To this we add

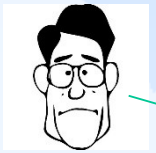
CrowdSourcing

- *People are often not privacy aware*
- *So we propose an aggregation/crowdsourcing component*
- *People with different privacy attitudes share data with different policies*
- *Objective Risk posed by particular items?*

To find that, we use psychometrics

- *Psychometric functions are applied in different fields*
- *In education they are used most widely in the famous SAT, GRE, GMAT tests in the US*
- *Given enough data, a question's "objective" difficulty is measured according to the ability of the test takers (and vice versa)*

Item Response Theory (IRT)



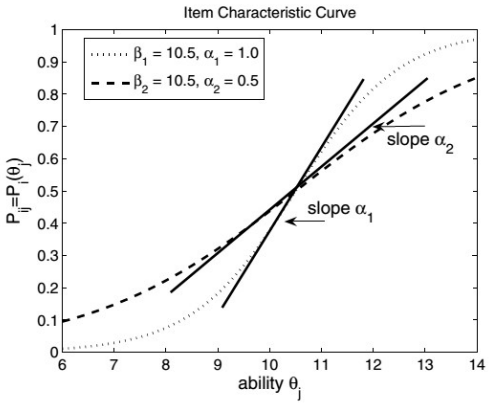
Examinees with different **ability** traits

Examinees' **responses**



Question with **difficulty** property

Correct *Reponse Probability*
=1/1+ e↑−(ability−difficulty)





Users' **sharing behavior**

Users of various privacy **attitudes**







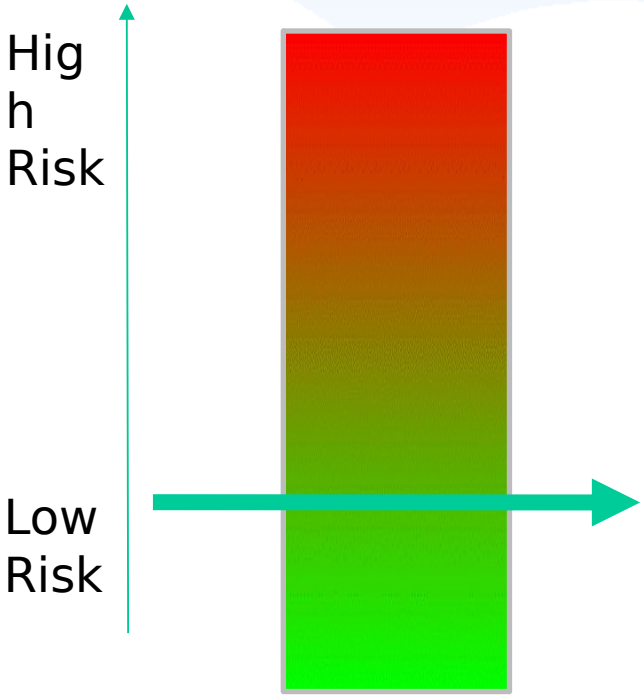
Item with **sensitivity** property

Disclosure *Likelihood*
=1/1+ e↑−(attitude−sensitivity)

Using this - Risk Meter

Automatic risk detection (and presentation)

- *Show the current risk and allow adjusting while visualizing the result*

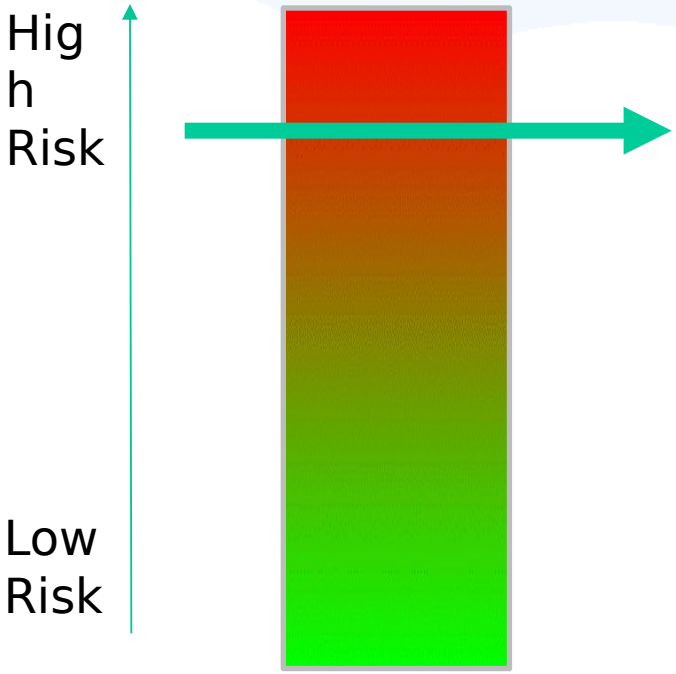


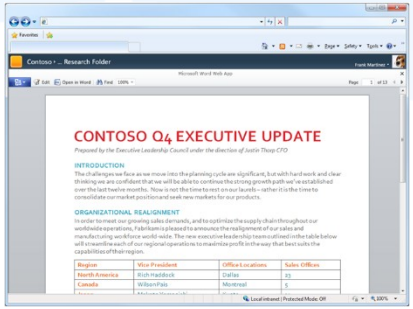
Encrypted Storage



2- Risk Meter

Show the current risk and allow adjusting while visualizing the result







Main Message!

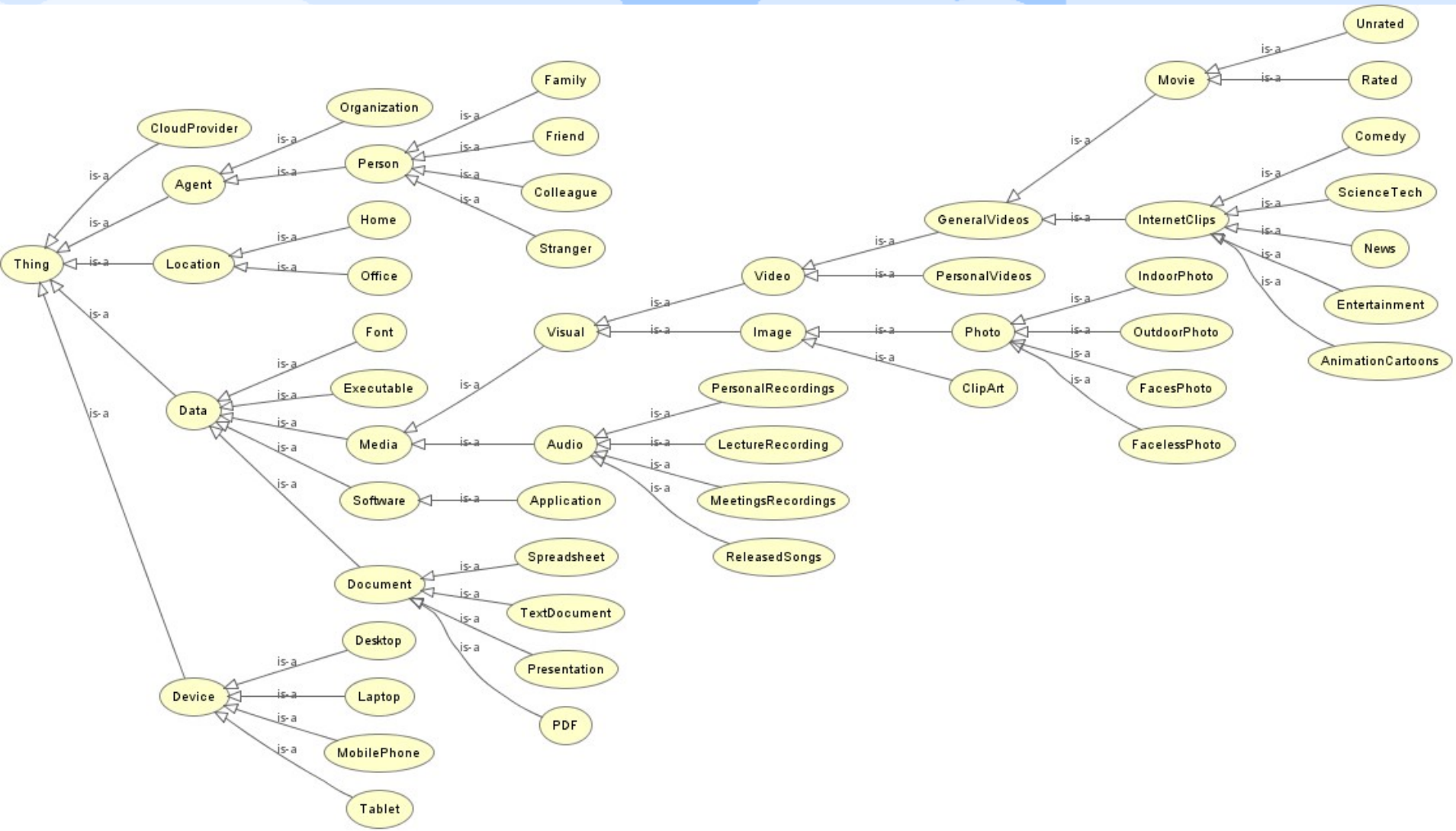
- *Context*
- *Crowd Sourcing*
- *Psychometrics*
- *Recommendation (not control)*

References

[1] Liu, Kun, and Evimaria Terzi. "A framework for computing the privacy scores of users in online social networks." *ACM Transactions on Knowledge Discovery from Data (TKDD)* 5.1 (2010): 6.

[2] Pallapa, Gautham, Nirmalya Roy, and Sajal K. Das. "A scheme for quantizing privacy in context-aware ubiquitous computing." *Intelligent Environments, 2008 IET 4th International Conference on*. IET, 2008.

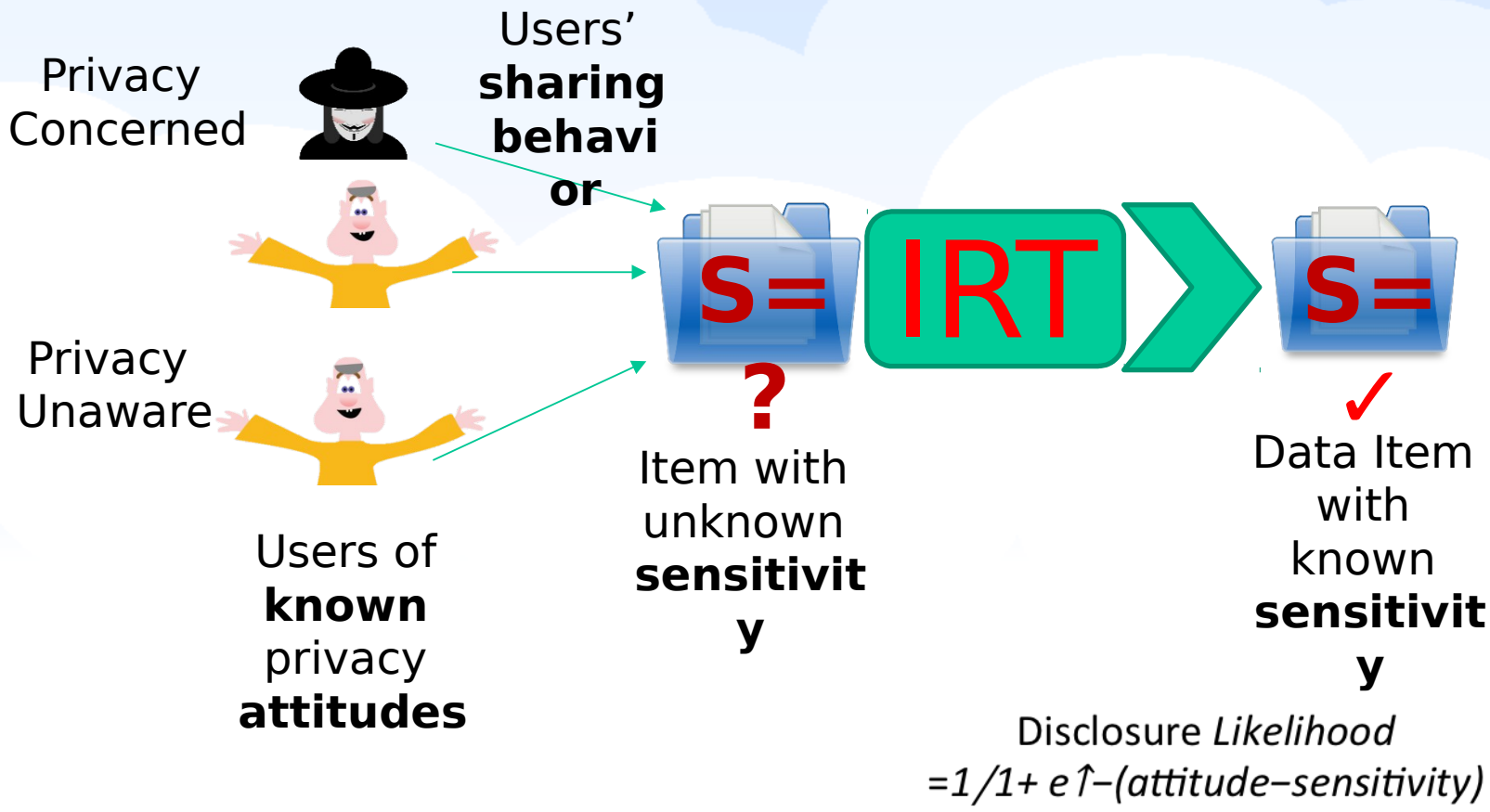
Example Vocabulary for the personal cloud



Privacy Aware Design

- *Minimizing the personal information sent to the cloud*
- *Protecting personal information in the cloud*
- *Maximizing user control*
- *Specifying the limit and purpose of data usage by concentrating on context*

Applying IRT in Dynamic Systems



Applying IRT in Dynamic Systems

